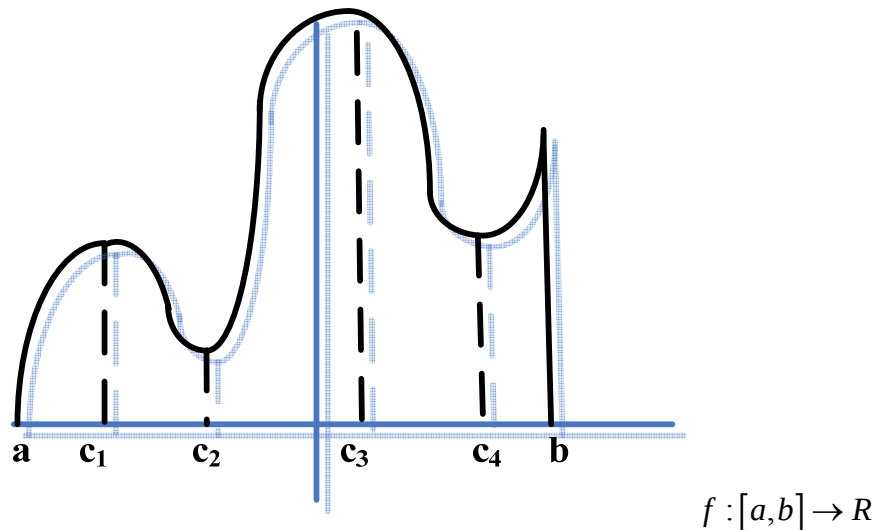


§4-1 Maximum and Minimum Values

Homework : 5,20,26,37,41,45,52,53,59,63,70,74,75

- Absolute (Global) extreme
- Local (Relative) extreme



Absolute Maximum : $f(c_3)$; f has an absolute max. at $x = c_3$.

Absolute Minimum : $f(a)$.

Local min : $f(a), f(c_2), f(c_4)$.

Local max : $f(b), f(c_1), f(c_3)$.

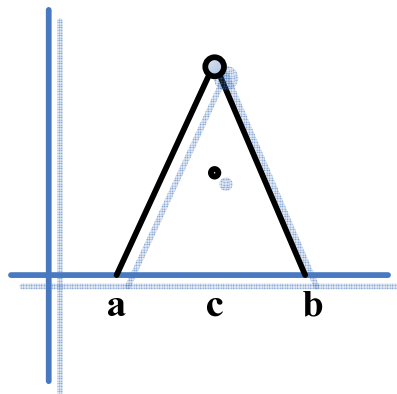
Note that $f(c_4) > f(c_1)$.

Theorem :

f is continuous on $[a, b] \Rightarrow f$ attains it's absolute extrema.

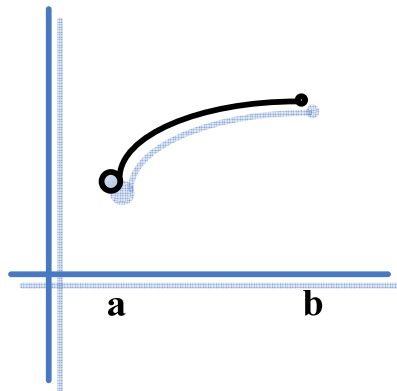
- Assumptions of the theorem need to be satisfied.
Otherwise the assertion of the theorem fails.

Example :



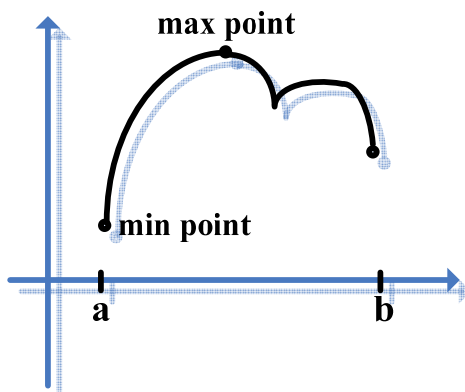
- not continuous at $x = c$
- no absolute max.

Example :



- not continuous at (a, b)
- no absolute min.

Example :



Theorem :

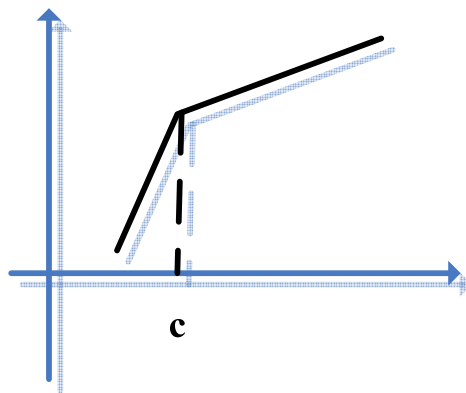
f has a local extremum at $x = c$.

$\Rightarrow f'(c) = 0$ or $f'(c)$ does not exist.

- The converse of the theorem is not necessarily true.

Example :

$f(x) = x^3$, $f'(0) = 0$, $f(0)$ is neither a local min nor a local max.

Example :

- $f'(c)$ does not exist
- $f(c)$ is neither a local min nor a local max.

Definition :

c : a critical point (c.p.) of f

$\Leftrightarrow f'(c) = 0$ or $f'(c)$ does not exist.

Remark :

Extreme take place at critical points.

- A critical point is not necessarily a local extremum point.

Steps to find absolute extrema for a continuous function defined on $[a, b]$.

- (1) Find all c.p.'s.
- (2) Compare the value of f at all c.p.'s and endpoint points.

x	c.p's	a	b
y			

Example 1 : $f(x) = x^3 + 3x^2 + 1, \quad -\frac{1}{2} \leq x \leq 4.$

Solution : $f'(x) = 3x^2 + 6x = 3x(x + 2)$

x	$-\frac{1}{2}$	0	4
y	$\frac{13}{8}$	1	113

Answer : $\begin{cases} \text{maximum} = 113 \\ \text{minimum} = 1. \end{cases}$

Example 2 : Let $a, b > 0.$

$f(x) = x^a(1-x)^b, 0 \leq x \leq 1.$

Solution : $f'(x) = ax^{a-1}(1-x)^b - bx^a(1-x)^{b-1}$

$$= x^{a-1}(1-x)^{b-1} [a(1-x) - bx]$$

$$= x^{a-1}(1-x)^{b-1} [a - (a+b)x]$$

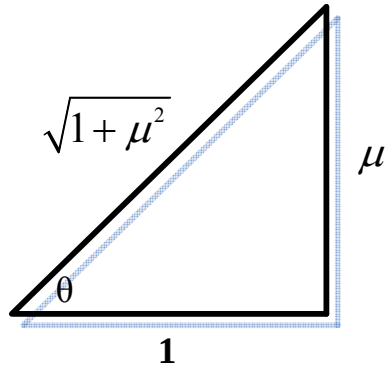
x	0	$\frac{a}{a+b}$	1
y	0	$\left(\frac{a}{a+b}\right)^a \left(\frac{b}{a+b}\right)^b$	0

Answer : $\begin{cases} \text{maximum} = \left(\frac{a}{a+b}\right)^a \left(\frac{b}{a+b}\right)^b = \frac{a^a b^b}{(a+b)^{a+b}} \\ \text{minimum} = 0. \end{cases}$

Example 3 : $F(\theta) = \frac{\mu\omega}{\mu \sin \theta + \cos \theta} \quad 0 \leq \theta \leq \frac{\pi}{2}, \mu, \omega > 0$

Solution : $F'(\theta) = \frac{-\mu\omega(\mu \cos \theta - \sin \theta)}{(\mu \sin \theta + \cos \theta)^2} = 0$

$$\Rightarrow \mu = \tan \theta$$



θ	0	$\frac{\pi}{2}$	$\tan^{-1} \mu$
$F(\theta)$	$\mu\omega$	ω	$\frac{\mu\omega}{\sqrt{1 + \mu^2}}$

$$\Rightarrow F \text{ is minimized when } \tan \theta = \mu.$$